

Lecture 04 - Probability Foundations and Random Variables

This lecture introduces the probability language needed to reason about uncertainty in finance and business. Lecture 3 summarized observed data. Lecture 4 moves one step further: it asks how we can describe situations whose outcomes are not yet known. Future demand, investment returns, project revenues, customer churn, loan defaults, and market losses are all uncertain before they are observed. Probability gives us a disciplined way to name possible outcomes, assign numerical weights to them, combine events, condition on information, and eventually summarize uncertain numerical quantities.

The lecture has two connected parts. The first part introduces probability foundations: outcomes, sample spaces, events, complements, unions, intersections, mutually exclusive events, conditional probability, and independence. The second part introduces random variables: numerical functions of uncertain outcomes. Random variables allow us to compute expected values, variances, and standard deviations of uncertain quantities such as revenue, profit, return, loss, or number of defaults.

1. Probability as a Language for Uncertainty

Probability is not only a calculation tool. It is a language for describing uncertainty precisely.

In business and finance, many decisions must be made before the relevant facts are known. We do not know next year's sales, next month's market return, whether a borrower will default, whether a product launch will succeed, or whether a customer will churn. Probability allows us to represent those unknowns in a structured way.

The basic workflow is:

1. Define what can happen.
2. Assign probabilities to the possible outcomes.
3. Define events that answer the question of interest.
4. Use probability rules to combine events or update probabilities when new information is known.

This workflow makes assumptions visible. It also helps avoid vague statements such as “this project is risky” or “this event is unlikely.” Instead, we can ask what the possible outcomes are, how much probability each receives, and how those probabilities change when information changes.

2. Outcomes and Sample Space

An outcome is one possible result of an uncertain situation. The sample space is the full set of possible outcomes included in the model. It is often denoted by S or Ω .

For example, the sample space for one roll of a standard die is

$$S = \{1, 2, 3, 4, 5, 6\}.$$

For a product launch, a simple sample space might be

$$S = \{\text{Failure, Moderate success, High success}\}.$$

The outcomes in a sample space must satisfy two conditions. They must be mutually exclusive, meaning that only one outcome can occur. They must also be collectively exhaustive, meaning that one of the listed outcomes must occur.

These conditions matter. If outcomes overlap, we may double count. If outcomes are not exhaustive, some possible cases are missing from the model.

In simple textbook examples, the sample space is often obvious. A coin has two sides. A die has six faces.

In business and finance, however, defining the sample space is already a modelling decision. One analyst may classify a product launch into three outcomes: failure, moderate success, and high success. Another analyst may use five outcomes or define success using revenue thresholds. Neither choice is automatic. The sample space depends on the problem, the available information, and the level of detail required.

3. Assigning Probabilities

Once the sample space is defined, probabilities are assigned to the possible outcomes.

A probability is a number between 0 and 1:

$$0 \leq P(\text{outcome}) \leq 1.$$

The probabilities assigned to all possible outcomes in the sample space must sum to 1:

$$\sum_i p_i = 1.$$

For a fair die,

$$P(1) = P(2) = \dots = P(6) = \frac{1}{6}.$$

In business applications, probabilities may come from several sources. They may come from a theoretical model, from historical frequencies, or from expert judgment. For example, probabilities for a product launch may be based on market research, previous launches, statistical modelling, or managerial assessment.

This means that probabilities are often model inputs, not directly observed facts. Two analysts can agree on the same sample space but disagree about the probabilities assigned to each outcome. Their conclusions may then differ because their probability models differ.

4. Events

An event is a subset of the sample space. It is a collection of outcomes that answer a question of interest. Suppose the product launch sample space is

$$S = \{\text{Failure, Moderate success, High success}\}.$$

The event “the launch is successful” can be defined as

$$A = \{\text{Moderate success, High success}\}.$$

If

$$P(\text{Moderate success}) = 0.50$$

and

$$P(\text{High success}) = 0.20,$$

then

$$P(A) = 0.50 + 0.20 = 0.70.$$

This example illustrates an important difference between outcomes and events. The outcomes in a sample space are the elementary possibilities. An event may group several outcomes together. We often care about events because business questions are usually broader than exact outcomes. We may not only ask whether the launch has “moderate success”; we may ask whether it succeeds at all.

5. Complements, Unions, and Intersections

If A is an event, its complement A^c is the event that A does not occur. The complement rule is

$$P(A^c) = 1 - P(A).$$

For example, if A is “the customer buys the product,” then A^c is “the customer does not buy the product.” If A is “the return is between -2 percent and 3 percent,” then A^c is “the return is less than -2 percent or greater than 3 percent.”

Events can also be combined. The intersection $A \cap B$ is the event where both A and B occur. The union $A \cup B$ is the event where at least one of the two events occurs: A , B , or both.

Complements, unions, and intersections are regions of a sample space

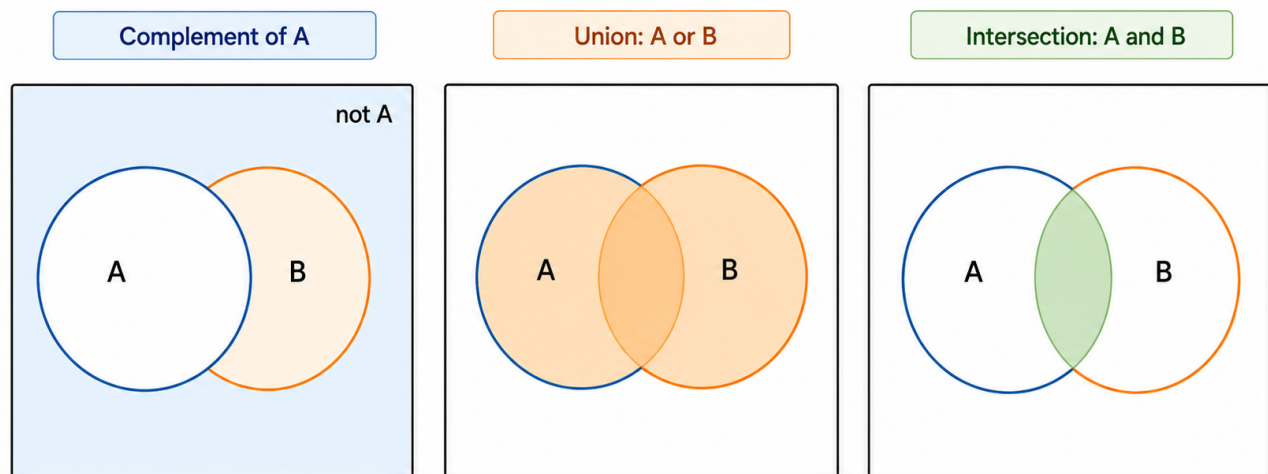


Figure 1. Complements, unions, and intersections are different regions of a sample space.

The union formula is

$$P(A \cup B) = P(A) + P(B) - P(A \cap B).$$

The subtraction is necessary because the overlap $A \cap B$ is counted twice when $P(A)$ and $P(B)$ are added. The formula removes that double counting.

This is especially important because outcomes must be mutually exclusive, but events do not have to be. For example, in a product launch analysis, one event may be “commercially successful” and another

may be “strategically valuable.” A launch can be both, so the events overlap.

6. Mutually Exclusive Events

Two events are mutually exclusive if they cannot happen at the same time. In set notation,

$$A \cap B = \emptyset.$$

Therefore,

$$P(A \cap B) = 0.$$

For mutually exclusive events, the union formula simplifies to

$$P(A \cup B) = P(A) + P(B).$$

Consider a return R and the following events:

$$A = \{R < -2\%\},$$

$$B = \{R > 3\%\},$$

$$C = \{R > 0\},$$

and

$$D = \{|R| < 3\%\} = \{-3\% < R < 3\%\}.$$

Events A and B are mutually exclusive because a return cannot be below -2 percent and above 3 percent at the same time. Events A and C are also mutually exclusive because A requires a negative return while C requires a positive return. But B and C are not mutually exclusive: every return above 3 percent is also positive. Likewise, A and D overlap between -3 percent and -2 percent, and C and D overlap between 0 and 3 percent.

The practical method is to translate the events into regions and ask whether any possible outcome belongs to both.

Independent vs mutually exclusive

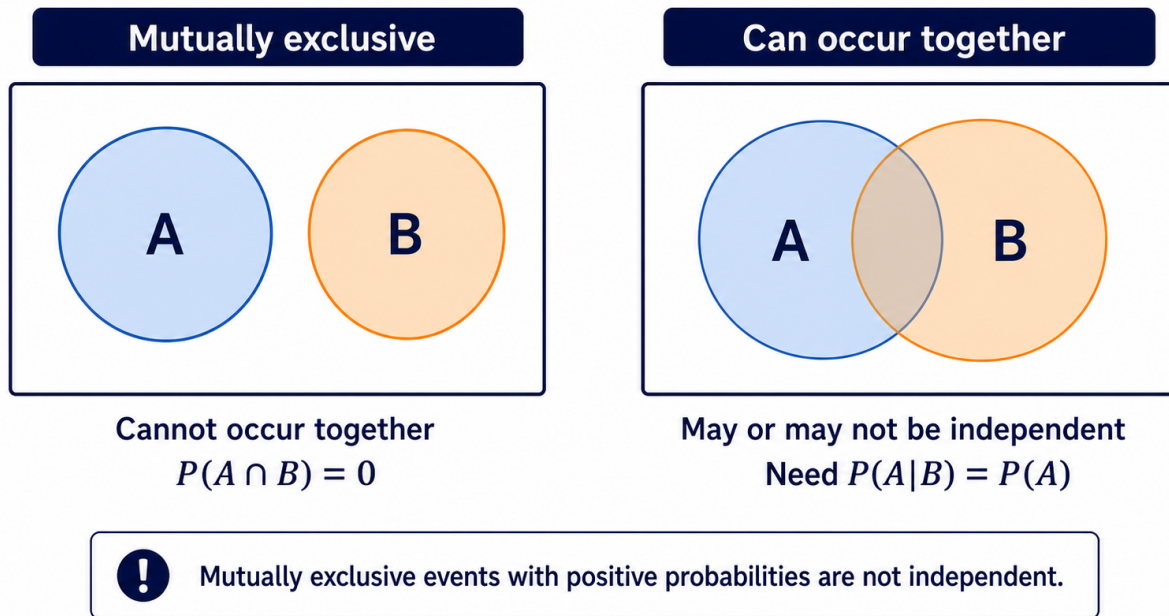


Figure 2. Mutually exclusive events cannot occur together; independence is a different concept.

The figure highlights a common confusion. Mutually exclusive events with positive probabilities are not independent. If A occurs, then B cannot occur, so learning that A occurred changes the probability of B .

7. Conditional Probability

Conditional probability is the probability of one event after we have learned that another event occurred. We write

$$P(A | B)$$

and read it as “the probability of A given B .” If $P(B) > 0$, then

$$P(A | B) = \frac{P(A \cap B)}{P(B)}.$$

The denominator is the key to the interpretation. When we condition on B , we restrict attention to the part of the sample space where B occurs. Among those cases, we ask how often A also occurs.

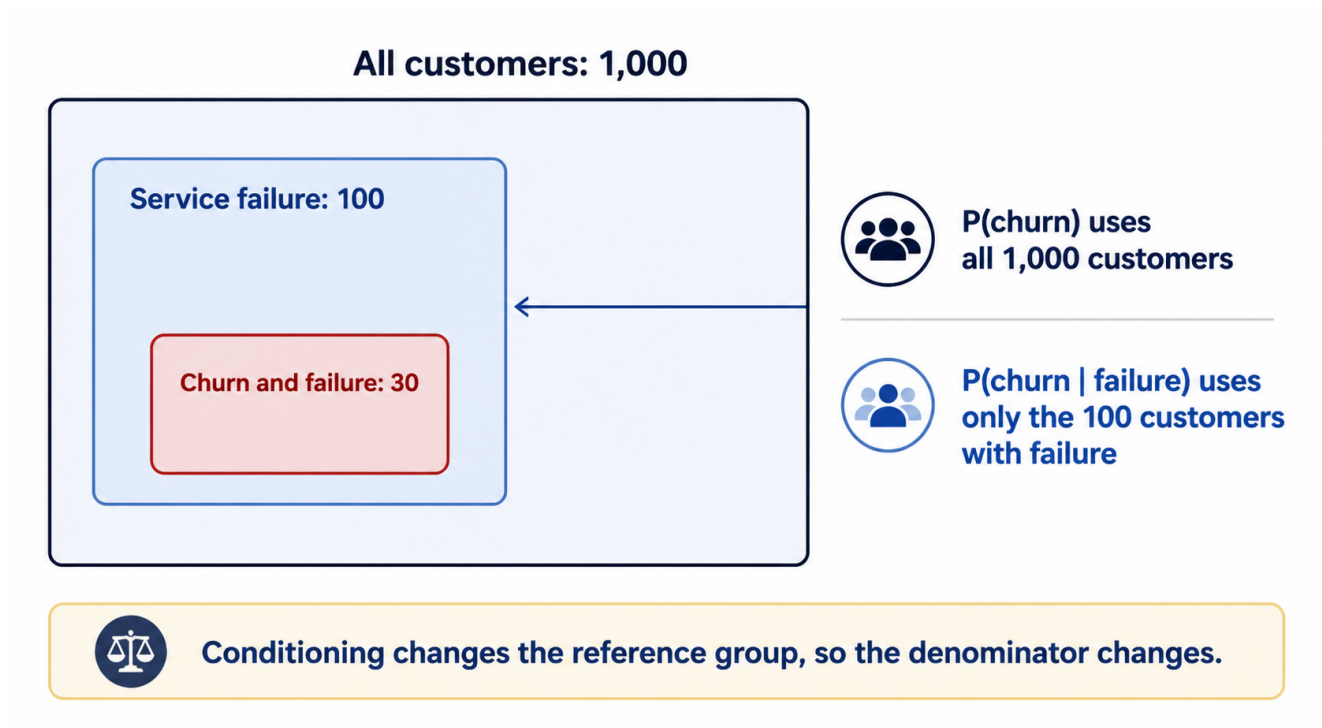


Figure 3. Conditioning changes the reference group, so the denominator changes.

For example, suppose we study customer churn and service failure. If there are 1,000 customers, 100 had a service failure, and 30 both churned and had a service failure, then

$$P(\text{churn} \mid \text{failure}) = \frac{30}{100} = 0.30.$$

The denominator is 100, not 1,000, because we are conditioning on service failure. We are no longer asking about all customers. We are asking about the subset of customers with service failure.

This is the idea behind many business questions. The probability of default given a high-risk flag is not the same as the probability of being flagged given default. The probability of a stock loss given a market downturn is not the same as the unconditional probability of a stock loss. Conditioning changes the reference group.

8. Conditional Probability from a Two-Way Table

Conditional probabilities are often easiest to compute from a table.

Suppose a portfolio contains 42 firms classified by region and sector:

Region	Telecom	Banking	Total
Europe	5	12	17
US	18	7	25
Total	23	19	42

If a firm is selected at random, the unconditional probability that it belongs to the banking sector is

$$P(\text{Banking}) = \frac{19}{42} \approx 45.2\%.$$

Now suppose we know that the selected firm is from the US. The probability that it belongs to banking becomes

$$P(\text{Banking} \mid \text{US}) = \frac{7}{25} = 28\%.$$

The denominator changed from 42 firms to 25 firms because the information “US” restricts the reference group to the US row.

The same result follows from the formula:

$$P(\text{Banking} \mid \text{US}) = \frac{P(\text{Banking} \cap \text{US})}{P(\text{US})} = \frac{7/42}{25/42} = \frac{7}{25}.$$

Knowing the region changes the probability assigned to the sector. This leads directly to the idea of independence.

9. Independence

Two events A and B are independent if knowing that one occurred does not change the probability of the other.

One way to write this is

$$P(A \mid B) = P(A).$$

Equivalently,

$$P(A \cap B) = P(A)P(B).$$

These two definitions are connected. Starting from conditional probability,

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)}.$$

If $P(A \mid B) = P(A)$, then

$$P(A \cap B) = P(A)P(B).$$

In the portfolio table above, the event “the company is in banking” and the event “the company is from the US” are not independent, because knowing that the firm is from the US changes the banking probability from about 45 percent to 28 percent.

Independence is often a strong assumption in finance. Events can be connected through common risk factors such as market conditions, sector exposure, interest rates, credit conditions, liquidity, or macroeconomic news. When we assume independence, we should justify it or check whether the data support it.

10. From Outcomes to Random Variables

Probability begins with outcomes and events. Many finance questions, however, require numerical quantities. We care about revenue, profit, return, loss, number of defaults, or portfolio value. A random variable creates the bridge from qualitative uncertainty to quantitative analysis.

A random variable assigns a numerical value to each outcome in the sample space. If s_i is an outcome, then a random variable X assigns a value

$$X(s_i) = x_i.$$

For the product launch example, the outcomes may be failure, moderate success, and high success. Define X as the revenue generated by the project:

Scenario	Probability	Revenue X
Failure	0.30	0
Moderate success	0.50	50,000
High success	0.20	150,000

Here X is a discrete random variable because it can take a finite list of values.

The uncertainty has not disappeared. The launch will still end in one of the scenarios. But now the scenarios have been translated into numbers, so we can compute expected revenue, variance, standard deviation, downside probabilities, and other summaries.

11. Probability Distribution of a Discrete Random Variable

A probability distribution of a discrete random variable lists the possible values of the variable and the probabilities attached to those values.

If X can take values $x_1, \dots, x_k$, with probabilities $p_1, \dots, p_k$, the probabilities must satisfy

$$0 \leq p_i \leq 1$$

and

$$\sum_{i=1}^k p_i = 1.$$

It is useful to distinguish outcomes from values of the random variable. Different outcomes can lead to the same value of X . In that case, the probability of a value of X is the sum of the probabilities of all outcomes that produce that value.

Consider three fair coin tosses. The sample space contains eight outcomes:

$$\{HHH, HHT, HTH, HTT, THH, THT, TTH, TTT\}.$$

Each outcome has probability $1/8$. Define two random variables on this same sample space:

$$X = 1 \text{ point for each Head and } -1 \text{ point for each Tail,}$$

and

$$Y = 10 \text{ points if all three tosses are equal, and } 0 \text{ otherwise.}$$

The same sample space now produces two different random variables.

Outcome	Probability	X	Y
HHH	1/8	3	10
HHT	1/8	1	0
HTH	1/8	1	0
HTT	1/8	-1	0
THH	1/8	1	0
THT	1/8	-1	0
TTH	1/8	-1	0
TTT	1/8	-3	10

The probability distribution of X is:

Value of X	Probability
-3	1/8
-1	3/8
1	3/8
3	1/8

The probability distribution of Y is:

Value of Y	Probability
0	6/8 = 3/4
10	2/8 = 1/4

This example is important because it shows that the same outcomes can generate different numerical questions. The random variable determines what is being measured.

12. Expected Value

The expected value of a random variable is its probability-weighted average. For a discrete random variable X with possible values x_i and probabilities p_i ,

$$E[X] = \sum_i p_i x_i.$$

Using the product launch example,

$$E[X] = 0.30(0) + 0.50(50,000) + 0.20(150,000) = 55,000.$$

The expected value is also called the mean of the random variable and is sometimes denoted by μ .

The interpretation is crucial. The expected value is not a forecast that one launch will generate exactly 55,000. In the model, the possible revenues are 0, 50,000, and 150,000. The expected value summarizes the center of the probability distribution. If many comparable launches were repeated under the same probability model, the long-run average revenue would tend toward the expected value.

This is similar to a sample mean, but the weights are different. In a dataset, the mean uses observed frequencies. For a random variable, expected value uses probabilities from a probability model.

13. Variance and Standard Deviation of a Random Variable

Expected value describes the center of a probability distribution, but it does not describe risk by itself. Two projects can have the same expected revenue and very different dispersion. One project may produce values close to the expected value; another may have large upside and large downside.

The variance of a discrete random variable measures the probability-weighted squared distance from the expected value:

$$\text{Var}(X) = \sum_i p_i (x_i - E[X])^2.$$

The standard deviation is the square root of the variance:

$$\sigma_X = \sqrt{\text{Var}(X)}.$$

These definitions are similar to the dataset variance and standard deviation from Lecture 3. The difference is that for a random variable we use probability weights, not sample frequencies. The center is $E[X]$, not the sample mean $\bar{x}$.

14. Expected Value and Variance: Coin-Toss Example

Return to the random variable X from the three fair coin tosses:

Value of X	Probability
-3	1/8
-1	3/8
1	3/8
3	1/8

The expected value is

$$E[X] = \frac{1}{8}(-3) + \frac{3}{8}(-1) + \frac{3}{8}(1) + \frac{1}{8}(3) = 0.$$

Because $E[X] = 0$, the variance is

$$\text{Var}(X) = \frac{1}{8}(-3 - 0)^2 + \frac{3}{8}(-1 - 0)^2 + \frac{3}{8}(1 - 0)^2 + \frac{1}{8}(3 - 0)^2.$$

Therefore,

$$\text{Var}(X) = \frac{1}{8}(9) + \frac{3}{8}(1) + \frac{3}{8}(1) + \frac{1}{8}(9) = \frac{24}{8} = 3.$$

The standard deviation is

$$\sigma_X = \sqrt{3} \approx 1.73.$$

In words, the expected score is zero, but typical outcomes are dispersed around that center by about 1.73 points. The expected value tells us the center; the standard deviation tells us how variable the possible outcomes are.

15. Historical Frequencies and Probability Models

In applied work, probabilities are often estimated from historical frequencies. Using the stress threshold in the Lecture 4 data, 33 out of 180 trading days are market-stress days, so one possible estimate of the probability of market stress is $33/180 \approx 18.3\%$.

This is a useful bridge between descriptive statistics and probability. Descriptive statistics summarize what happened in the observed sample. A probability model uses those summaries as estimates for what may happen in an uncertain future situation.

The bridge requires an assumption. Historical frequencies can be used as probability estimates only if the historical sample is relevant for the future question. If market conditions change, if the sample is too short, or if the future process differs from the past, then the probability model may be misleading.

This is why careful language matters. A statement such as “large stock losses occurred on 36 of 180 days in this sample” is descriptive. A statement such as “the probability of a large stock loss is 20 percent” is a model-based statement. The second statement may be useful, but it requires the assumption that the sample is informative about the probability we want to model.

16. Careful Probability Interpretation

Probability statements should not be interpreted as certainties. If a default probability is 3 percent, default is unlikely under the model, but it can still happen. If the expected profit is 44, the realized profit does not have to be 44. If two events occur together in one observation, that does not prove they are dependent. If two events can occur together, that does not prove they are independent.

The safest interpretation always names the model, the event, and the reference group. For conditional probability, name the denominator. For expected value, say that it is a probability-weighted average, not a guaranteed outcome. For independence, say whether the probability of one event changes when we condition on the other.

This lecture provides the language needed for the next steps in the course. Distributions, sampling, confidence intervals, hypothesis tests, and regressions all rely on the same foundation: outcomes, probabilities, events, conditioning, random variables, expected values, and dispersion.